Why conflicts

Multi-patterning starts by finding which features cannot share the same mask

The idea

- GOLDEN: 6 nodes, 5 edges (A-B, A-C, B-C, D-E, E-F), min_same_mask=2

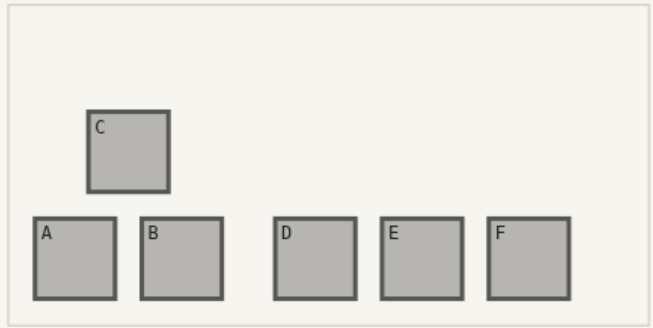
Features

CONFLICT GRAPH

Features

Step 1 / 5 · feat · module01-01-conflict-graph

n=6



Explanation

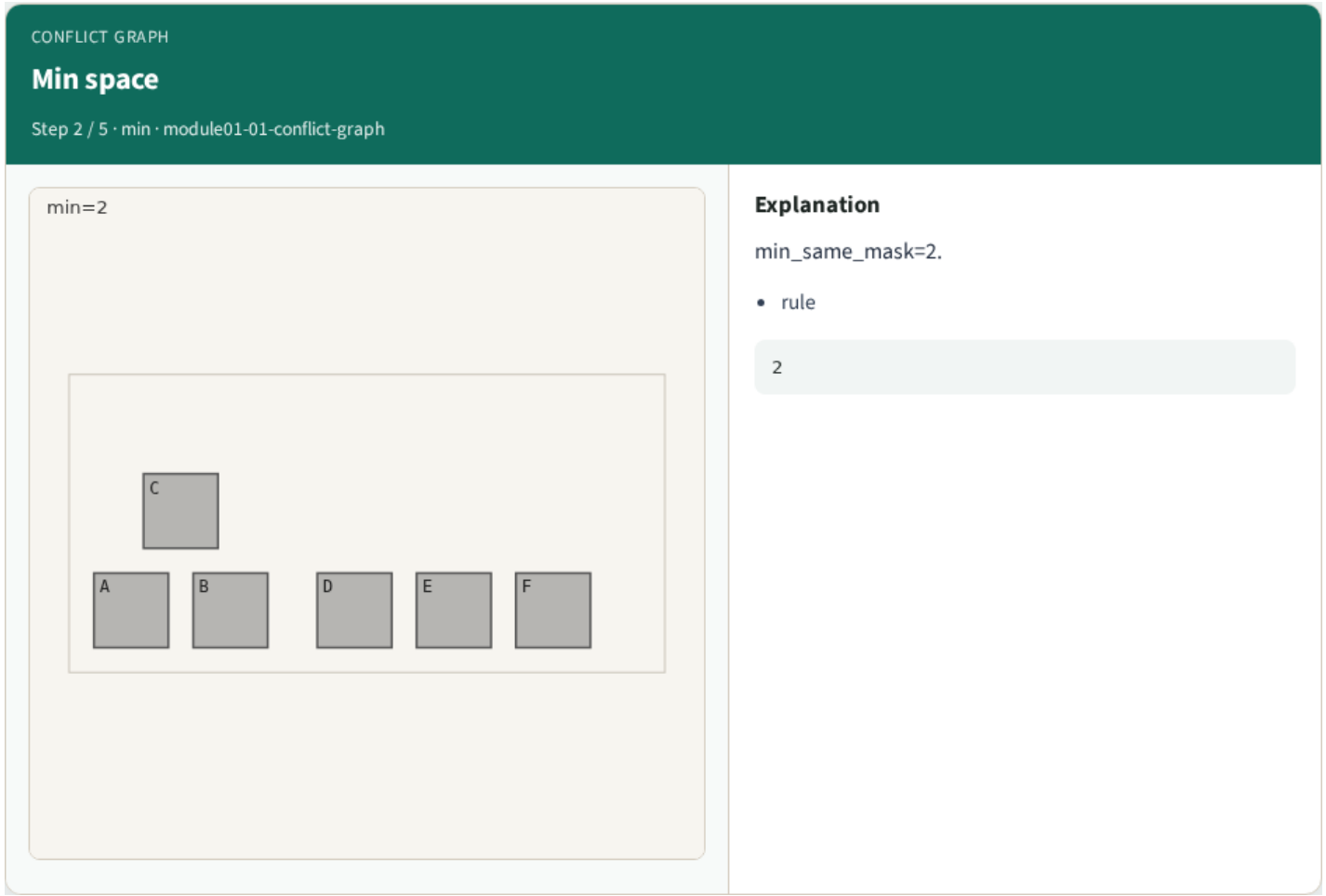
Six features A–F on the toy.

- A–F

n=6

Features

Min space



Min space

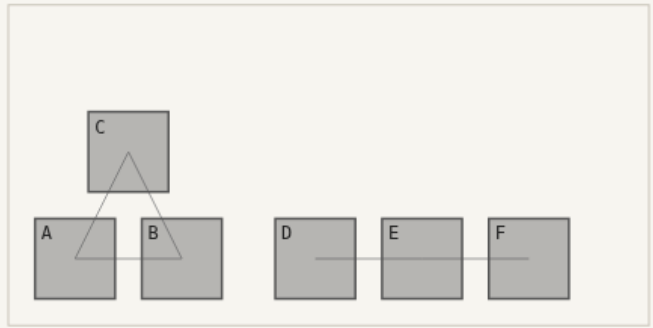
Edges

CONFLICT GRAPH

Edges

Step 3 / 5 · edges · module01-01-conflict-graph

edges=5



Explanation

Conflict if gap < 2 → 5 edges.

- graph

```
E=5
```

Edges

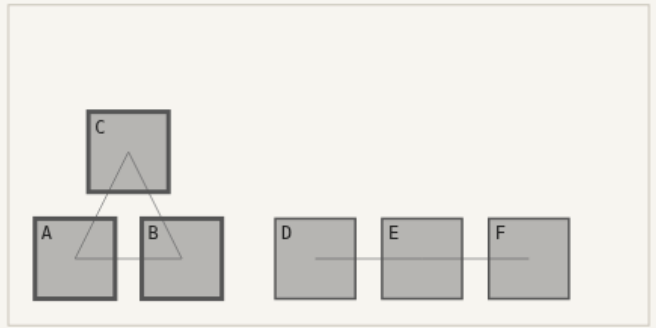
Triangle

CONFLICT GRAPH

Triangle

Step 4 / 5 · tri · module01-01-conflict-graph

triangle



Explanation

A–B–C form a triangle.

- odd

```
ABC
```

Triangle

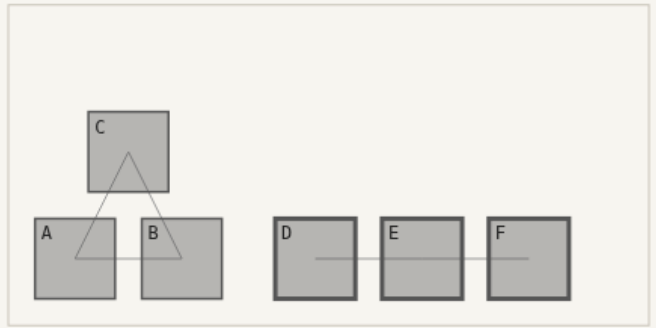
Path

CONFLICT GRAPH

Path

Step 5 / 5 · path · module01-01-conflict-graph

path



Explanation

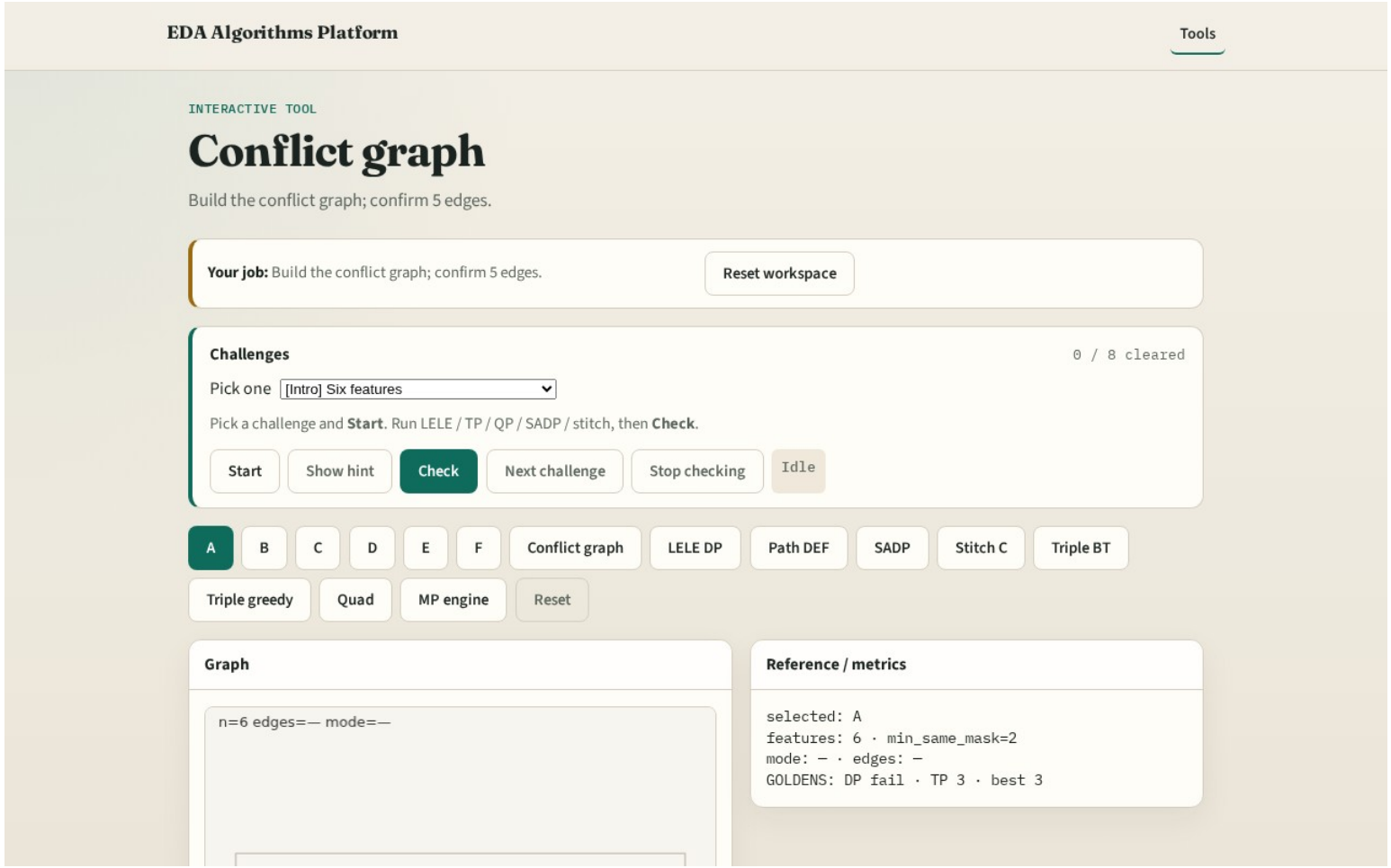
D–E–F is a path.

- bipartite piece

```
DEF
```

Path

Browser lab track



Browser lab starter

Implement track

- Call build_conflict_graph(); assert feature_count==6 and edge_count==5

Pitfalls

- Using center distance instead of bbox gap
- Forgetting min_same_mask from JSON

Your turn

- Next: LELE double patterning as 2-coloring